

## Part 1: Phonons and Fermionic Gas

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# Black Body

Paul Haskins

March 27, 2025

Black body implies a gas of phonons, bosonic particles.

1. Relativistic  $\omega_c(1, \epsilon)$  2.  $N$  not conserved,  $\mu = 0$  grand canonical is approx canonical and  $J$  approx is  $F$ .

The total average energy density is given by

$$\langle \epsilon \rangle = \frac{\langle E \rangle}{V} = \frac{8\pi^5}{15c^3 h^3} \beta^{-4} \quad (1)$$

$$\langle \epsilon \rangle = \int_0^\infty d\epsilon \frac{\omega_c(1, \epsilon)}{V} \epsilon \langle n(\epsilon) \rangle \quad (2)$$

We could define

$$E(\epsilon) d\epsilon = \frac{\omega_c}{V} \epsilon \langle n(\epsilon) \rangle_{\mu=0} d\epsilon \quad (3)$$

Above is the average energy distribution of photons.

Let us express it in terms of the frequency  $\omega$ . Using  $\epsilon = \hbar\omega$  we have

$$E(\omega) d\omega = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta\hbar\omega} - 1} \quad (4)$$

$$\implies_{x=\beta\hbar\omega} E(x) \propto \frac{x^3}{e^x - 1} \quad (5)$$

$$x_0 = \beta_0 \hbar \omega_0$$

$\implies \omega_0 = \frac{kT_0}{\hbar}$ . This determines the max of the distribution determined by

$$\left. \frac{dE(x)}{dx} \right|_{x_0} = 0 \quad (6)$$

We can examine limits.

$E(x) \approx x^3 e^x$  for  $x \gg_1$ . QM

$E(x) \approx x^2$  for  $x \ll_1$  classical.

This is known as Wien's law. UV-catastrophe.

The only difference between photons and phonons is the factor of the degeneracy where photons have a deg of 2 while phonons have 3N as they have mass.

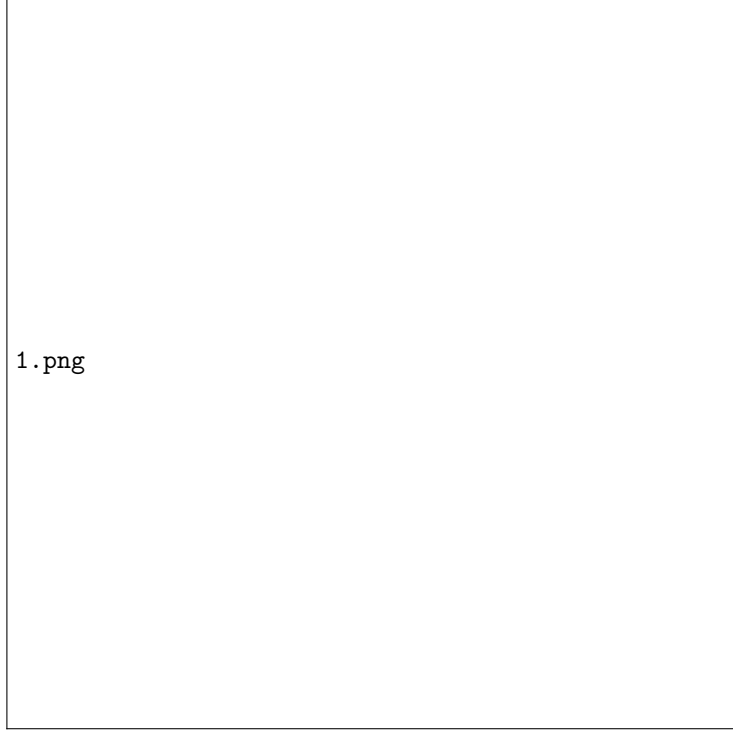


Figure 1: Planck's law. The monotonic dashed line is the classical case.

## Non-Relativistic fermionic gas

$$\ln(\Xi) = \sum_{n=1}^{\infty} \ln(1 + ze^{-\beta\epsilon_n}) \quad (7)$$

This holds for  $0 < z < \infty$

$$= \ln(1 + z) + \int_0^{\infty} d\epsilon \omega_c \ln(1 + ze^{-\beta\epsilon}) \quad (8)$$

Previously we had

$$\omega_c d\epsilon = (\deg) \frac{V}{h^3} (2m)^{\frac{3}{2}} 2\pi \epsilon^{\frac{1}{2}} d\epsilon \quad (9)$$

Using IBP and letting  $x = \beta\epsilon$

$$\ln(\Xi) = \ln(1 + z) - (\dots) \int_0^{\infty} \frac{dx x^{\frac{3}{2}}}{1 + \frac{e^x}{z}} \quad (10)$$

Using analytical cont for  $-z$  and the def of polylogarithm.  
Our integral is  $\mathcal{L}_{i\frac{5}{2}}(-z)$

$$\mathcal{L}_{i\frac{5}{2}}(-z) = -\frac{1}{\Gamma(r)} \int_0^\infty \frac{dt t^{(r-1)}}{1 + \frac{e^t}{z}} \quad (11)$$

which converges for  $z \geq -1$

$$\ln \Xi = -\ln(1+z) (\deg) \left( \frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{3}{2}} \mathcal{L}_{i\frac{5}{2}}(-z) \quad (12)$$

holds for  $0 \leq z \leq \infty$

\* There is no bound on  $z$  and hence on  $\mathcal{L}_{i\frac{5}{2}}$ . Thus there is no special physics associated with the ground state.

\* \* The polylogarithm term dominates for both large and small  $z$  and combining with \* we can safely neglect the ground state contribution.

$$\ln \Xi = -(\deg) \left( \frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{3}{2}} \mathcal{L}_{i\frac{5}{2}}(-z) \quad (13)$$

$$\langle N \rangle = z \frac{d \ln \Xi}{dz} = (\deg) \left( \frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{3}{2}} \mathcal{L}_{i\frac{3}{2}}(-z) \quad (14)$$

$$\langle E \rangle = -\frac{d \ln \Xi}{d\beta} \Big|_{v,z} = -\frac{3}{2} (\deg) \left( \frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{5}{2}} \mathcal{L}_{i\frac{3}{2}}(-z) = \frac{3}{2} PV \quad (16)$$

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$$\mathcal{L}_{i\frac{5}{2}}(-z) \approx -z \left[ 1 - \frac{z}{2^r} + \dots \right] \quad (17)$$

for  $z \ll_1$  and for  $z \gg 1$ .

$$\mathcal{L}_{i\frac{5}{2}}(-z) \approx -\frac{(\ln(z))^r}{\Gamma(r+1)} \left[ 1 + \frac{\pi^2 r(r-1)}{(\ln(z))^2} \right] \quad (18)$$

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But also

$$\langle E \rangle = \frac{3}{2\beta} \langle N \rangle \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} = \begin{cases} \frac{\langle N \rangle}{\beta} \ln(z) & \text{if } z \gg_1 \\ \frac{3}{2} kT \langle N \rangle & \text{if } z \ll_1 \end{cases} \quad (20)$$

$$F = \langle E \rangle - TS = \mu \langle N \rangle + J = \langle N \rangle kT \left[ \ln(z) - \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} \right] \quad (21)$$

where in the last equality  $\beta\mu = \ln(z)$

$$S = \frac{1}{T} (\langle E \rangle - F) = \langle N \rangle k \left[ \frac{5}{2} \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} \right] \quad (22)$$

$$C_{v,n} = T \frac{\partial S}{\partial T} |_{V,N} = T \frac{\partial S}{\partial z} |_{V,N} \frac{\partial z}{\partial T} |_{V,N} \quad (23)$$

what is  $\frac{\partial z}{\partial T}$

Set  $N$  fixed  $= < N > \implies \frac{\partial N}{\partial T} |_{V,N} = 0$

$$0 = (deg) V \left( \frac{2\pi m k}{h^2} \right)^{\frac{3}{2}} \left[ -\frac{3}{2} T^{\frac{1}{2}} \mathcal{L}_{i\frac{3}{2}} - T^{\frac{3}{2}} \frac{\partial}{\partial T} \mathcal{L}_{i\frac{5}{2}} \right] \quad (24)$$

$$-\frac{3}{2} T^{\frac{1}{2}} \mathcal{L}_{i\frac{3}{2}} = -T^{\frac{3}{2}} \frac{\mathcal{L}_{i\frac{5}{2}}}{-z} \frac{\partial z}{\partial T} \quad (25)$$

$$\frac{1}{z} \frac{\partial z}{\partial T} = -\frac{3}{2T} \frac{\mathcal{L}_{i\frac{3}{2}}(-z)}{\mathcal{L}_{i\frac{1}{2}}(-z)} \quad (26)$$

Sub and compute.

$$C_{V,N} = -\frac{9}{4} N K \frac{\mathcal{L}_{i\frac{3}{2}}(-z)}{\mathcal{L}_{i\frac{1}{2}}(-z)} + \frac{15}{4} N k \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} \quad (27)$$

Classically  $C_{V,N} \approx \frac{3}{2} N k$  as expected the leading behavior of  $\mathcal{L}_{i_r}$  for  $z \ll 1$ .

**Exercises:** Show that in the canonical regime  $S$  behaves as expected.

$$\mathcal{L}_{i\frac{3}{2}}(-z) \approx -\frac{(\ln z)^{\frac{3}{2}}}{\Gamma(\frac{5}{2})} = -\frac{< N >}{V} \left[ \left( \frac{h^2}{2\pi m} \right)^{\frac{3}{2}} \beta^{\frac{3}{2}} \frac{1}{(deg)} \right] \quad (28)$$

In last equals we solve in terms of  $< N >$ .

$$\implies \mu(z \gg 1) \approx \left( \frac{N}{V} \right)^{\frac{2}{3}} \quad (29)$$

which is fixed and independent of  $T$ .

As  $T \rightarrow 0$ ,  $\mu(T \rightarrow 0) \approx \exists_F$

What does this mean for  $< n(\epsilon) >?$

Under the limit  $T \rightarrow 0$

$$< n(\epsilon) > = \frac{1}{e^{\beta(\epsilon - \mu)} + 1} = \begin{cases} 1 & \text{if } \epsilon < \epsilon_f \\ 0 & \text{if } \epsilon > \epsilon_f \end{cases}$$

(31)

$$N = \int_0^\infty d\epsilon \omega_c < n(e) > \quad (32)$$

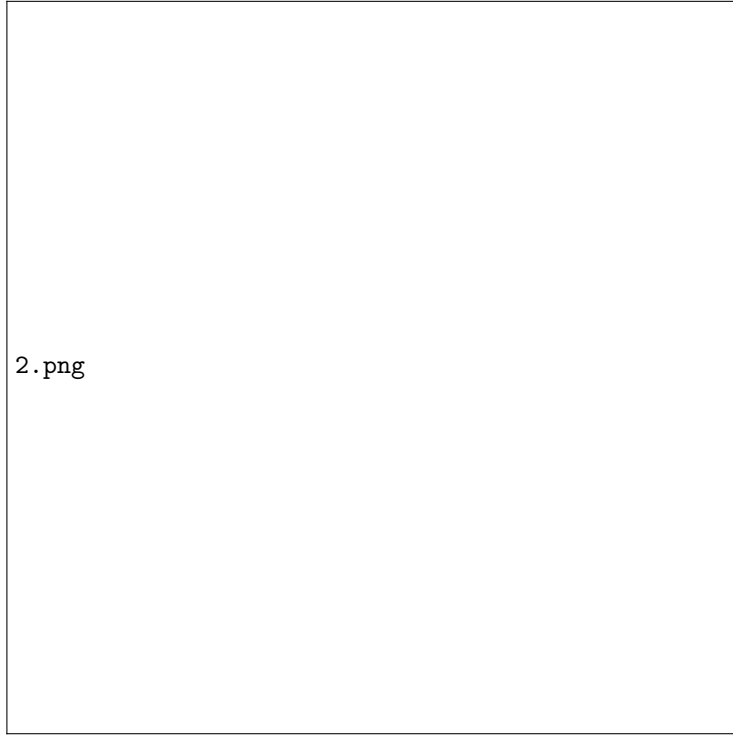


Figure 2: 2.png

As  $T \rightarrow 0$  with  $z \gg 1$ .

$$\Rightarrow \int_0^{\epsilon_F} d\epsilon \omega_c = \epsilon_F^{\frac{3}{2}} \quad (33)$$

$$\langle E \rangle = \int_0^\infty d\epsilon \epsilon \langle n(\epsilon) \rangle = \int_0^{\epsilon_F} d\epsilon \epsilon \omega_c = \epsilon_F^{\frac{5}{2}} \quad (34)$$

$$\Rightarrow \langle E \rangle = \frac{3}{5} N \epsilon_F \quad (35)$$

Now let us study  $C_v$  for  $\beta \gg 1$  ( $z \gg 1$ ) Substituting the leading polylogarithms behavior in  $C_V$  results to zero! We need to keep higher order terms in the second expansion in the box 18

Doing so

$$\Rightarrow C_v \approx \frac{\pi^2}{2} N k \left( \frac{kT}{\epsilon_F} \right) \quad (36)$$

we can tell what system (bosonic or fermionic) we are dealing with by the small T behavior of  $C_V$

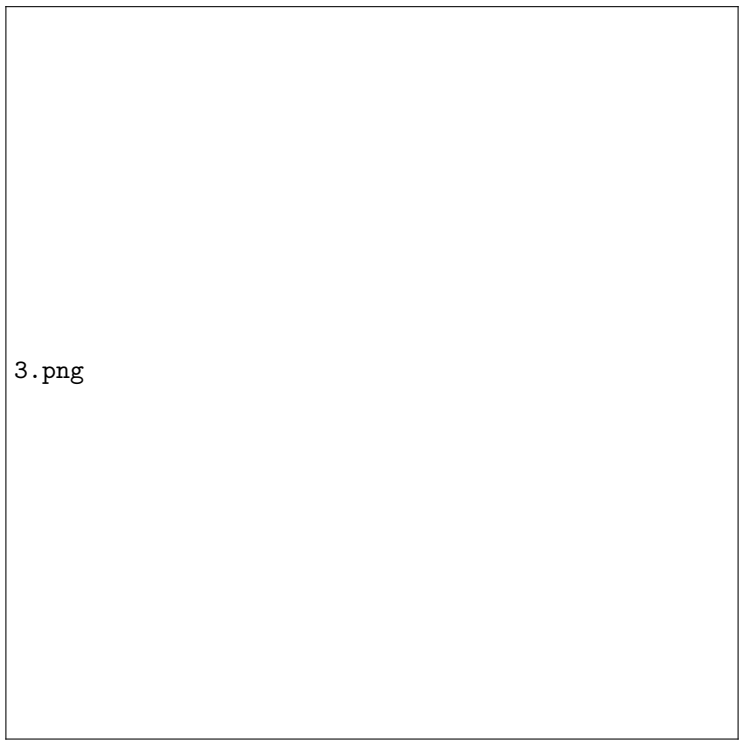


Figure 3:

$$T_F \approx \frac{\epsilon_F}{K} \tag{37}$$

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